Services Knowledge Base (SKB)

Image Filter – Inline vs. Offline
The influence of the Implicit Filter

Detailed explanation

FOR INTERNAL USE ONLY

NOT FOR DISTRIBUTION TO END CUSTOMERS

Image Filter Inline and Offline

1. Inline Image Filer

To apply an inline “Image Filter” the user has to select the following option under the protocol parameters:

Resolution > Filter Image > Image Filter (Image 01)

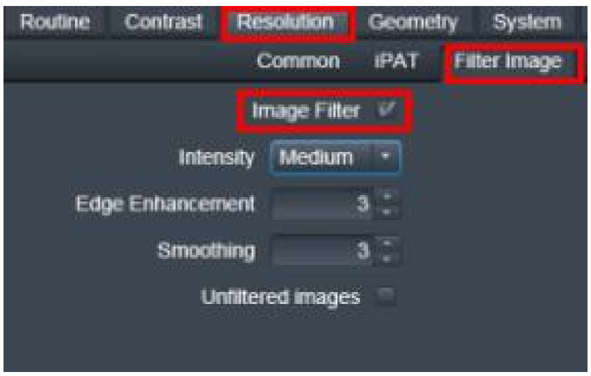

Image 01. Inline Image Filter

There are three different "**Intensity**" values for the “Image Filter” (Image 02):

- Sharp - the effect of this filter is the "weakest" one
- Medium - the effect of this filter is "medium"
- Smooth - the effect of this filter is the "strongest" one

Image 02. Inline Image Filter – There are 3 different intensities of an Image Filter: Sharp, Medium or Smooth.

Each of these filters (Sharp, Medium or Smooth) can be tuned up - i.e. refined - by using two additional parameters: the “Edge Enhancement” and the “Smoothing” factors (Image 03).

Image 03. Inline Image Filter - With the "Edge Enhancement" and the "Smoothing" values the user can refine the filter selected under "Intensity". Possible values of these parameters: from 1 to 5 (3 is the default value)

Here you can find the details of the possible values of these two additional parameters:

1. "**Edge Enhancement**"
   - It controls the enhancement of the structures – borders or regions with 2 very different signal intensities
   - The corresponding "Intensity" will be modified accordingly to the next values:
     - 3 – default value: no change in the selected "Intensity" regarding enhancement of the structures
     - 4 – higher enhancement effects of the structures
     - 5 – the highest enhancement effects of the structures
     - 2 – less enhancement effects of the structures
     - 1 – the lowest enhancement effects of the structures

2. "**Smoothing**"
   - it controls the noise reduction
   - The corresponding "Intensity" will be modified accordingly to the next values:
     - 3 – default value: no change in the selected "Intensity" regarding smoothing effects in noisy regions
     - 4 – higher smoothing effects in noisy regions
     - 5 – the highest smoothing effects in noisy regions
     - 2 – less smoothing effect in noisy regions
     - 1 – the lowest smoothing effect in noisy regions

2. Offline Image Filter

To apply an offline "Image Filter" the user has to select the following option from the Patient Browser or from the Viewing Task Card:

Evaluation > Dynamic Analysis > Image Filter (Image 04)

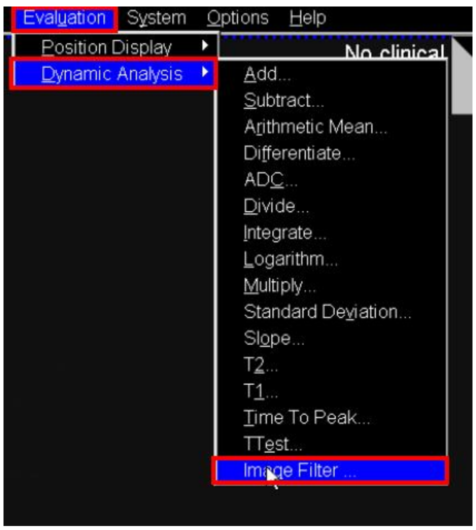

Image 04. Offline Image Filter

The same parameters defined in the previous section - "Intensity", "Edge Enhancement" and "Smoothing" values - can be selected when applying the Offline filter (Image 05)

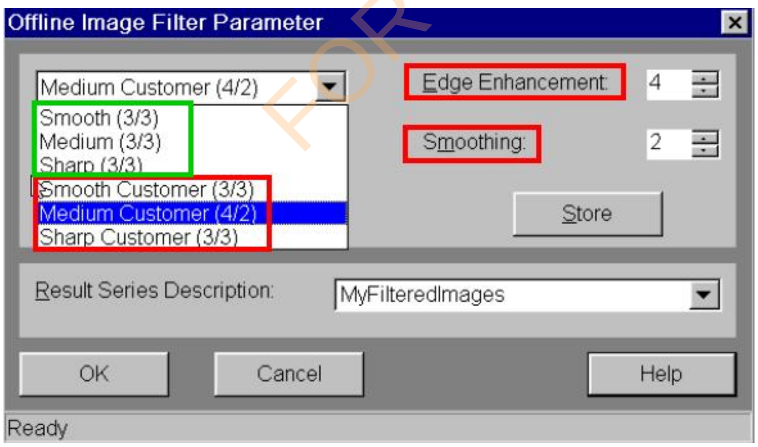

Image 05. Offline Image Filter - "Smooth (3/3)", "Medium (3/3)" and "Sharp (3/3)" are the default settings of the filters. If different "Edge Enhancement" or/and "Smoothing" factors are wanted, the user must select one of the "Customer" filters and adjust the corresponding values. The name of the filter will automatically be updated with the "Edge Enhancement" and "Smoothing" selected factors - in this example "Medium Customer (4/2)"

Example 1 - effect of the parameters of Image Filter

The user selects the following "Image Filter":

- “Intensity"=Medium
- "Edge Enhancement"=5
- "Smoothing"=1

As a result:

- The enhancement effect of the structures would be higher than the default Medium filter and
- The smoothing effect of the noisy regions would be lower than the default Medium filter

For comparison of the different results please watch the attachments of this SKB entry: "Unfiltered_Sharp_Medium_Smooth.avi" and "Medium33_Medium51.avi" (images from a pineapple)

3. Image Filter can be applied only once

If the images have been already filtered with an "Image Filter" - it does not matter whether Inline or Offline - the resulted filtered images cannot be filtered again with any additional "Image Filter".

Therefore, if the user tries to apply an “Offlline Image Filter” on already filter images, the following message will pop-up (Image 06):

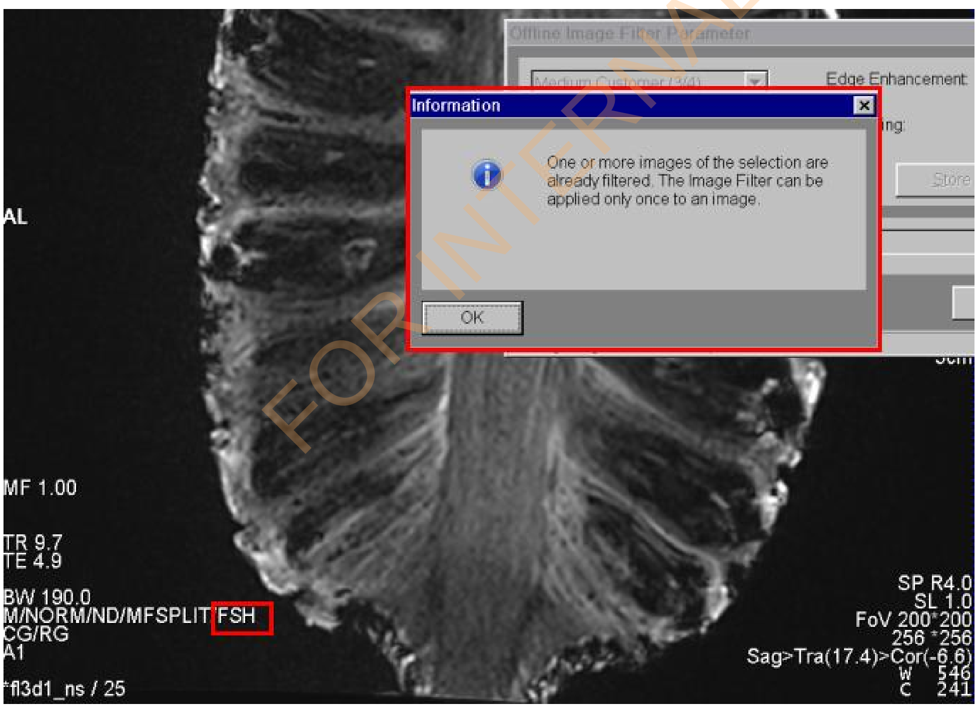

Image 06. Information message when the user tries to apply a second “Image Filter”. In this case the image had been already filtered with a default Sharp filter.

4. Labelled in the images

Depending on the selected “Image Filter”, the filtered images will usually get the following labels:

- **FSHx_y** - "Intensity"=Sharp - "Edge Enhancement"=x - "Smoothing"=y
- **FMx_y** - "Intensity"=Medium - "Edge Enhancement"=x - "Smoothing"=y
- **FSMx_y** - "Intensity"=Smooth - "Edge Enhancement"=x - "Smoothing"=y
- **FSH** - "Intensity"=Sharp - Default filter, i.e. "Edge Enhancement"=3 and "Smoothing"=3
- **FM** - "Intensity"=Medium - Default filter, i.e. "Edge Enhancement"=3 and "Smoothing"=3
- **FSM** - "Intensity"=Smooth - Default filter, i.e. "Edge Enhancement"=3 and "Smoothing"=3

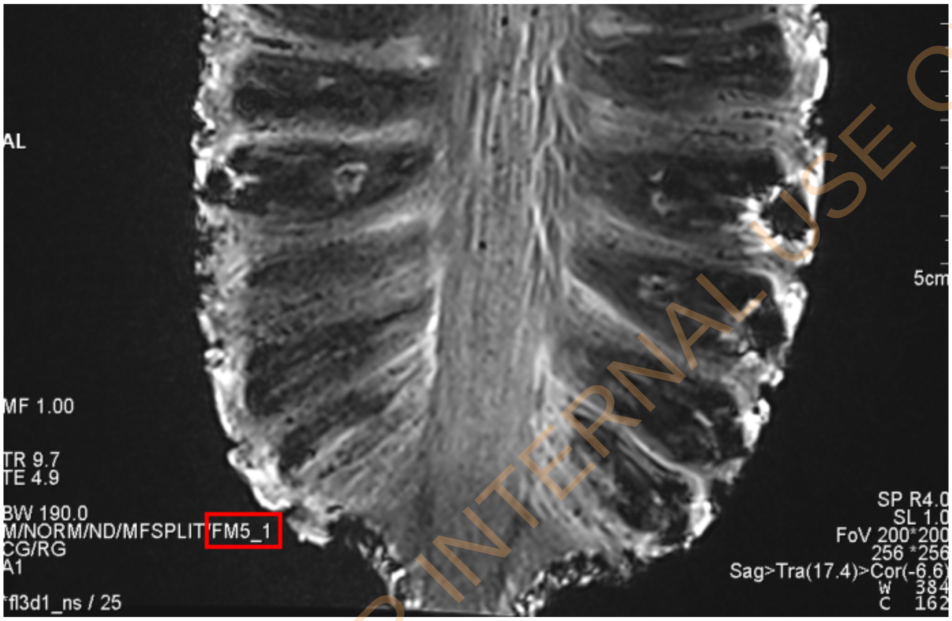

Image 07. Example of the resulted labelling after applying an Image Filter with "Intensity"=Medium - "Edge Enhancement"=5 - "Smoothing"=1.

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IMPLICIT FILTER

1. Definition

Since VB15A there is an additional "Implicit Filter" implemented in the systems. The implementation of this "Implicit Filter" uses the same parameters as defined for the previous "Image Filter", i.e. an "Intensity" value with "Edge Enhancement" and "Smoothing" factors.

These are the hardcoded parameters of the default "Implicit Filter":

- Intensity - Sharp
- Edge Enhancement - 3
- Smoothing - 3

That is, the default "Implicit Filter"is a FSH3_3.

These default values ("Intensity"=Sharp, "Edge Enhancement"=3 and "Smoothing"=3) cannot be directly modified by the user.

However, in software versions **before VE11A**, these values can be indirectly modified by defining an additional Inline "Image Filter" in the protocol. Here are the possible definitions of an "Implicit Filter":

- FSH3_3 – Default
- FSH3_4 – If the Inline "Image Filter" defined by the user is a FSH1_4 or FSH2_4
- FSH3_5 – If the Inline "Image Filter" defined by the user is a FSH1_5 or FSH2_5
- FSH4_3 – If the Inline "Image Filter" defined by the user is a FSH4_1 or FSH4_2
- FSH5_3 – If the Inline "Image Filter" defined by the user is a FSH4_1 or FSH4_2

From VE11A onwards, there is only one possible "Implicit Filter": the default one FSH3_3

è All the details about the potential modifications of the default "Implicit Filter" can be found in the section ***IMAGE FILTER IN COMBINATION WITH THE IMPLICIT FILTER***

2. Affected sequences

This "Implicit Filter", if enabled, is only applied to the following sequences:

- fl3d_vibe (VIBE)
- se
- tse - but not with triggered measurements, e.g. cardio-applications excluded
- haste - but not with triggered measurements, e.g. cardio-applications excluded
- tse_vfl (SPACE)

3. Labelled in the images

When an “Implicit Filter” is applied:

- The resulting filtered images are not labelled – different behavior as the one described for the normal “Image Filter” (Image 07)
- This information is not stored in the corresponding DICOM tag. Therefore, if the user tries to apply an additional “Offline Image Filter”, the system will not recognize that a previous filter had been already applied and the images can be filtered again (at the end, 2 filters will be applied to the images)

4. Activation / deactivation

The "Implicit Filter" can be enabled / disabled under:

Local Service > Configuration > Measurement > Meas. Settings > Inline Image Filter (Image 08).

è the default set up is "enabled"

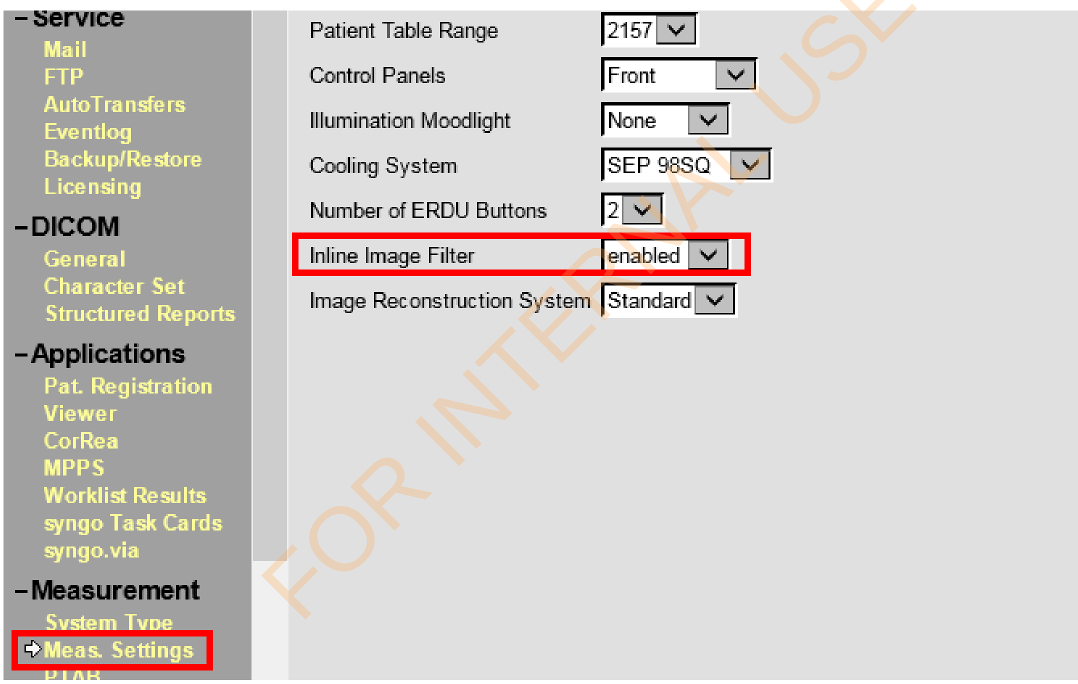

Image 08. "Implicit Filter" can be enabled/disabled under "Local Service > Configuration > Measurement > Meas. Settings > Inline Image Filter".

Notice that even if the name under “Local Service” is “Inline Image Filter”, the name used in this document to refer to this filter is “Implicit Filter”, whereas “Inline Image Filter” is used to refer to the inline application of the “Image Filter” under the protocol parameters: Resolution > Filter Image > Image Filter (Image 01)

IMAGE FILTER IN COMBINATION WITH THE IMPLICIT FILTER

This section ONLY applies to the combination of the IMPLICIT FILTER with an INLINE IMAGE FILTER selected by the user under *"Protocol parameters - Resolution > Filter Image > Image Filter"*.

Nevertheless, one can additionally find here:

- The reason why (sometimes) the resulting images after applying an Inline Image Filter are quite different from the resulting images after applying an Offline Image Filter (only for software versions before VE11A)
- The reason why (sometimes) when the user selects an Inline Image Filter with the additional option "Unfiltered images" (Image 09), only one series is provided as result instead of two series (only for software versions before VE11A)
- The reason why (sometimes) the filtered images are not labelled with the “Inline Image Filter” selected by the user (only for software versions before VE11A)

Image 09. The option to save the “Unfiltered images” can be additionally selected when using an “Inline Image Filter”.

1. Before VE11A (VB line and VD line)

Even if both filters are enabled, i.e. the Inline Image Filter and the Implicit Filter, the original images will be filtered only once.

- Default "Implicit Filter" (FSH3_3) is enabled
- User additionally selects an “Inline Image Filter”

The selection of which filter is applied (an "Implicit Filter" or the “Inline Image Filter") depends on the "strength" of the “Inline Image Filter” selected by the user.

1. ***If the user selects an Intensity = Medium or Smooth***
   - **The INLINE IMAGE FILTER will be always applied**, i.e. the applied filter will be the one selected by the user.

2. ***If the user selects an Intensity = Sharp*** à The application of an "Inline Image Filter" or an "Implicit Filter" will depend on the "Edge Enhancement" and "Smoothing" factors selected by the user:

   - **An IMPLICIT FILTER will be applied when the user selects an "Edge Enhancement" and/or "Smoothing" values lower than 3**. In these cases the value lower than 3 will be automatically set to 3. Here you can see the detailed list:
     - If the user defines a FSH1_4 or FSH2_4, the system will apply the **"Implicit Filter" FSH3_4**
     - If the user defines a FSH1_5 or FSH2_5, the system will apply the **"Implicit Filter" FSH3_5**
     - If the user defines a FSH4_1 or FSH4_2, the system will apply the **"Implicit Filter" FSH4_3**
     - If the user defines a FSH5_1 or FSH5_2, the system will apply the **"Implicit Filter" FSH5_3**
     - If the user defines a FSH1_1 or FSH1_2 or FSH1_3 or FSH2_1or FSH2_2 or FSH2_3, the system will apply the **"Implicit Filter" FSH3_3**

   - **The INLINE IMAGE FILTER will be applied when the user selects an "Edge Enhancement" and "Smoothing" values higher than 2**, i.e. the applied filter will be the one selected by the user. Notice that <u>if the user selects an "Inline Image Filter" FSH3_3, this filter is considered as an "Inline Image Filter" and not as "Implicit Filter"</u>

<u>Differences between the application of an "Implicit Filter" and the "Inline Image Filter"</u>

If an IMPLICIT FILTER is applied:

- The "Unfiltered images" will not be provided, even if the user additionally selected the option of "Unfiltered images" (Image 09). That is, the user will get only one series with the results of the "Implicit Filter"
- As described in one of the previous section, the resulting images will not be labelled; therefore there will not be any indication about the application of this filter.
- As previously mentioned, the information about the application of this filter will not be stored in the corresponding DICOM tag. Therefore, if the user tries to apply an additional Offline Image Filter, the system will not recognize that a previous filter had been already applied and the images can be filtered again.
  - > An additional "Offline Image Filter" is possible

If the INLINE IMAGE FILTER is applied:

- If the user selects the option of "Unfiltered images" (Image 09), the "Unfiltered images" will be provided.
  - > These "Unfiltered images" do not have any filter applied. Notice that this will not be the case in the software versions after VE11A (details are explained in the corresponding section).
- The images will be labelled as previously described (Image 07)
- The information about the application of this filter will be stored in the corresponding DICOM tag. Therefore, if the user tries to apply an additional Offline Image Filter, the system will recognize that a previous filter had been already applied.
  - > An additional "Offline Image Filter" on the resulted filtered images is not possible (Image 06)
  - > But an additional "Offline Image Filter" on the "Unfiltered images" would be possible.

Example 2 - applied filter when using Implicit Filter and Inline Image Filter

"Implicit Filter" is enabled and user selects an “Inline Image Filter" FSH5_1

> The system applies the following "Implicit Filter”: FSH5_3
> The "Unfiltered images", if selected, are not provided.
> Images are not labelled with the applied filter “FSH5_3”
> User can apply an additional “Offline Image Filter" afterwards if desired.

Example 3 - applied filter when using Implicit Filter and Inline Image Filter

"Implicit Filter" is enabled and user selects an “Inline Image Filter" FM5_1

> The system applies the selected "Inline Image Filter": FM5_1
> The "Unfiltered images", if selected, are provided.
> Images are labelled with the applied filter “FM5_1”
> The application of an additional “Offline Image Filter” on the resulted filtered images would not be possible (but it would be possible on the “Unfiltered images”).

Example 4 - applied filter when using Implicit Filter and Inline Image Filter

"Implicit Filter" is enabled and user selects an “Inline Image Filter" FSH2_2

> The system applies the following "Implicit Filter”: FSH3_3
> The "Unfiltered images", if selected, are not provided.
> Images are not labelled with the applied filter “FSH3_3”
> User can apply an additional “Offline Image Filter" afterwards if desired.

Inline Image Filter results vs. Offline Image Filter results

The already described behavior explains why in some cases the results from the “Inline Image Filter" are quite different from the results of the equivalent “Offline Image Filter" (for details, see Example 5 and Example 6).

Example 5 - different results when applying the Inline and Offline filters

The user compares the results of two scans:

§ The first one with an “Inline Image Filter” (Scan #1)
§ The second one without any “Inline Image Filter”, but the user will apply the corresponding “Offline Image Filter” afterwards (Scan #2)

Scan #1: "Implicit Filter" is enabled and user selects an “Inline Image Filter” FSM5_2

> The system applies the selected "Image Filter": FSM5_2
> The "Unfiltered images", if selected, are provided.
> Images are labelled with the applied filter “FSM5_2”

Scan #2: User repeats exactly the same Scan#1 but without selecting any “Inline Image Filter".

> The system applies the default "Implicit Filter": FSH3_3
> On the resulting images, the user applies the equivalent “Offline Image Filter” FSM5_2 (i.e. the same selection as in "Scan #1", since the expectations are to get the same results)
> Now the final resulting images are the ones after the application of the "Implicit Filter" (FSH3_3), which has been applied during the scan, and the application of the “Offline Image Filter" (FSM5_2)
   - Therefore, these images are quite different from the results of "Scan #1".
   - Notice that these results are also labelled with “FSM5_2”

Example 6 - different results when applying the Inline and Offline filters

The user compares the results of two scans:

§ The first one with an “Inline Image Filter” (Scan #1)
§ The second one without any “Inline Image Filter”, but the user will apply the corresponding “Offline Image Filter” afterwards (Scan #2)

Scan #1: "Implicit Filter" is enabled and user selects an “Inline Image Filter” FSH2_4

> The system applies the following "Implicit Filter": FSH3_4
> The "Unfiltered images", if selected, are not provided.
> Images are not labelled with the applied filter “FSH3_4” (and, of course, not with “FSH2_4”)

Scan #2: User repeats exactly the same Scan#1 but without selecting any “Inline Image Filter".

> The system applies the default "Implicit Filter": FSH3_3
> On the resulting images, the user applies the equivalent “Offline Image Filter” FSH2_4 (i.e. the same selection as in "Scan #1", since the expectations are to get the same results)
> Now the final resulting images are the ones after the application of the "Implicit Filter" (FSH3_3), which has been applied during the scan, and the application of the “Offline Image Filter" (FSH2_4)
   - Therefore, these images are quite different from the results of "Scan #1".
   - Notice that the offline results are now labelled with “FSH2_4”

2. After VE11A

If both filters are enabled, both filters are applied – i.e. the images are filtered twice.

- Default "Implicit Filter" (FSH3_3) is enabled
- User additionally selects an “Inline Image Filter”

In this case both filters, “Implicit filter” and “Inline Image Filter”, will be applied.

Therefore, after VE11A:

- The parameters of the "Implicit Filter" will never be modified. **"Implicit Filter" will always be FSH3_3**.
- First the “Implicit Filter” is applied. Afterwards the “Inline Image Filter” is applied.
- If the user selects the option of “Unfiltered images” (Image 09), the “Unfiltered images” will be provided.
  - > These “Unfiltered images” are the resulting images after the application of the “Implicit Filter” FSH3_3. That is, if “Implicit Filter” is enabled under the Local Service configuration, the “Unfiltered images” will always be filtered.
- The images will be labelled describing the use of the “Inline Image Filter” selected by the user (the label will not reflect the use of the “Implicit Filter”)
- The information about the application of the “Inline Image Filter” will be stored in the corresponding DICOM tag. Therefore, if the user tries to apply an additional Offline Image Filter, the system will recognize that a previous filter had been already applied.
  - > An additional “Offline Image Filter” on the resulted filtered images is not possible (Image 06)
  - > But an additional “Offline Image Filter” on the “Unfiltered images” would be possible.

Inline Image Filter results vs. Offline Image Filter results

Therefore, in these software versions, the results after the application of an “Inline Image Filter" are “almost” **(*)** the same as the results after applying the equivalent “Offline Image Filter”.

- > If the user compares the “Inline Image Filter” results (scanned with additionally the option “Unfiltered images”) with the results after applying the equivalent “Offline Image Filter” to these “Unfiltered images”, the results are “almost” **(*)** the same.
- > If the user repeats twice the same scan, once with “Inline Image Filter” and the other one without “Inline Image Filter”, and compares the results of the “Inline Image Filter” with the results after applying the equivalent “Offline Image Filter” to the images from the second scan, the results are “almost” **(*)** the same.

(*) Notice that if the user performs the subtraction of the “Inline Image Filter” results and the “Offline Image Filter” results, small differences between these results might be observed. The reason is that the “Inline Image Filter" is always applied to the images before storing them in DICOM format (floating numbers), whereas the “Offline Image Filter" is applied to the images already stored in DICOM format (integer numbers) – DICOM standard does not support floating numbers for images.